Year 10 Higher

REVISION PACK

Graphs 1

Name: _____

Class: _____

Graphs 1 · Gradients and equations of lines (Lessons 3.1, 3.3)

Worked example: a missing coordinate

A straight line passes through the points (1, 4) and (5, k). The gradient of the line is 2. Work out the value of k.

Step 1	Gradient formula	$\frac{k-4}{5-1}=2$
Step 2	Simplify the bottom	$\frac{k-4}{4}=2$
Step 3	Multiply by 4	$k-4=8$
Step 4	Add 4	$k=12$

Practice: a missing coordinate

1.1 A straight line passes through the points (3, 2) and $(7, k)$. The gradient of the line is 4. Work out the value of k.

1.2 A straight line passes through the points (1, 9) and $(4, k)$. The gradient of the line is −2. Work out the value of k.

1.3 A straight line passes through the points (−2, 5) and $(3, k)$. The gradient of the line is 3. Work out the value of k.

1.4 A straight line passes through the points (0, −3) and $(4, k)$. The gradient of the line is 2. Work out the value of k.

1.5 A straight line passes through the points (2, 1) and $(5, k)$. The gradient of the line is −3. Work out the value of k.

1.6 A straight line passes through the points (−1, 6) and $(3, k)$. The gradient of the line is 2. Work out the value of k.

Graphs 1 · Gradients and equations of lines (Lessons 3.1, 3.3)

Worked example: the equation of a line through two points

Find an equation of the straight line that passes through the points (−2, 9) and (2, −3).

Step 1	Gradient	$m = \dfrac{-3-9}{2-(-2)} = -3$
Step 2	Put in a point	$9 = -3 \times (-2) + c$
Step 3	Subtract 6 from both sides	$c = 3$
Step 4	Answer	$y = -3x + 3$

Practice: the equation of a line through two points

1.7 Find an equation of the straight line that passes through the points (2, 7) and (5, 19).

1.8 Find an equation of the straight line that passes through the points (−1, −1) and (3, 11).

1.9 Find an equation of the straight line that passes through the points (−1, 7) and (2, −2).

1.10 Find an equation of the straight line that passes through the points (1, 6) and (4, −6).

1.11 Find an equation of the straight line that passes through the points (−3, 10) and (1, 2).

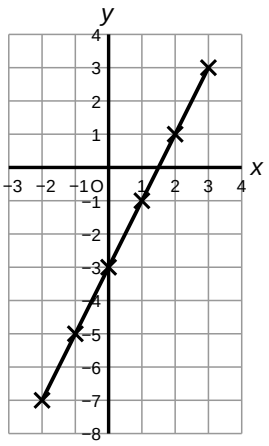
1.12 Find an equation of the straight line that passes through the points (0, −4) and (2, 6).

Graphs 1 · Drawing straight-line graphs (Lesson 3.2)

Worked example: a table of values and a graph

(a) Complete the table of values for $y = 2x - 3$ (b) Draw the graph of $y = 2x - 3$ for values of x from -2 to 3

x	-2	-1	0	1	2	3
y	-7	-5	-3	-1	1	3



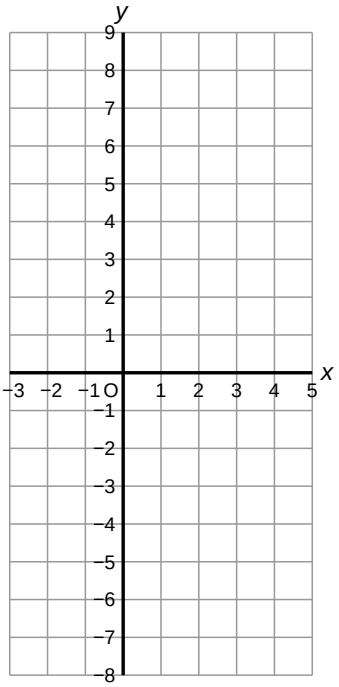
(a)	**Step 1**	Substitute each x	e.g. $x = -2$: $2 \times (-2) - 3 = -7$
	Step 2	Fill in the table	$-7, -5, -3, -1, 1, 3$
(b)	**Step 1**	Plot the points	$(-2, -7), (-1, -5), \ldots, (3, 3)$
	Step 2	Join with a ruler	one straight line from $x = -2$ to $x = 3$

Practice: a table of values and a graph

2.1 (a) Complete the table of values for $y = 3x - 1$

x	-2	-1	0	1	2	3
y						

(b) On the grid, draw the graph of $y = 3x - 1$ for values of x from -2 to 3

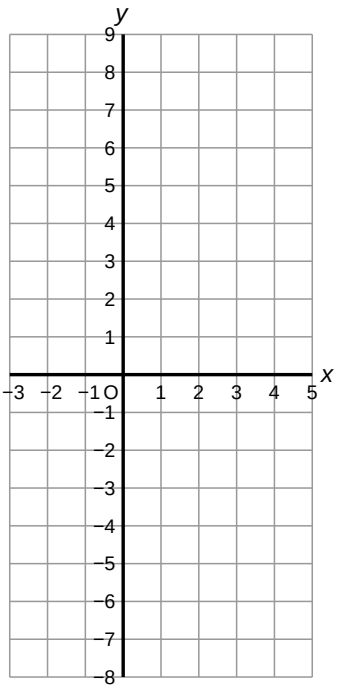


Practice: a table of values and a graph (continued)

2.2 (a) Complete the table of values for $y = -2x + 4$

x	-1	0	1	2	3	4
y						

(b) On the grid, draw the graph of $y = -2x + 4$ for values of x from -1 to 4

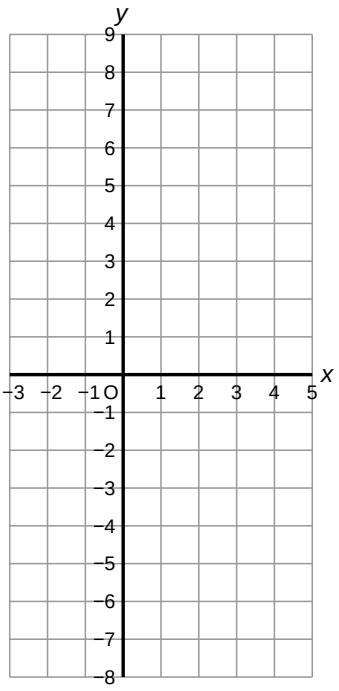


Practice: a table of values and a graph (continued)

2.3 (a) Complete the table of values for $y = \frac{1}{2}x + 2$

x	-2	-1	0	1	2	3	4
y							

(b) On the grid, draw the graph of $y = \frac{1}{2}x + 2$ for values of x from -2 to 4



Practice: a table of values and a graph (continued)

2.4 (a) Complete the table of values for $y = 4 - x$

x	-1	0	1	2	3	4	5
y							

(b) On the grid, draw the graph of $y = 4 - x$ for values of x from -1 to 5

